MTH101 - Calculus And Analytical Geometry

(Mid-Term) 2021 GRAND Quiz 100% Correct Solve (Alhamdulillah)

BY Virtual Assistance Sonu Ilyas Mughal

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MTH101 - Calculus And Analytical Geometry

MTH101:Grand Quiz — Quiz

Question # 1 of 30 (Start time: 05:18:43 PM, 21 June 2021)

Which of the following is equation of a line that passes through the point (5, -3) and whose slope is -1?

Select the correct option

- (●) y = - x + 2
- ( ) y = - x - 2
- ( ) y = x - 2
- ( ) y = x + 2

MTH101:Grand Quiz — Quiz Start Time: 05:18

Question # 2 of 30 (Start time: 05:19:35 PM, 21 June 2021)

$Which\ of\ the\ following\ is\ VERTEX\ of parabola\ represented\ by\ the\ equation : y = x^2 + 2$

Select the correct option — Reload

- ( ) None of these
- (●) (0, 2)
- ( ) (2, 6)
- ( ) (1, 3)

Click to Save Answer & Move to

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Question # 3 of 30 (Start time: 05:20:20 PM, 21 June 2021)

What is the derivative of $\sin(x^2) + \cos(2x)$?

Select the correct option

- ○ $2x\cos(x^2) + 2\sin 2x$
- ● $2x\cos(x^2) - 2\sin 2x$
- ○ *None of these.*
- ○ $2x\sin(x^2) + 2\cos 2x$

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Quiz Start

Question # 4 of 30 (Start time: 05:20:47 PM, 21 June 2021)

If $y = f(g(x))$ *then its derivative will be* . . . . . .

Select the correct option

- ○ $\frac{d}{dx}f(x).\ \frac{d}{dx}(g(x))$
- ● $\frac{d}{dx}f(g(x)).\ \frac{d}{dx}(g(x))$

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Question # 5 of 30 (Start time: 05:21:15 PM, 21 June 2021)

In xy-plane both x and y are positive in _____ quadrant.

Select the correct option

- First
- Second
- Forth
- Third

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Question # 6 of 30 (Start time: 05:21:45 PM, 21 June 2021)

$If\ f(x) = 2x\ and\ g(x) = x^2 + 3x,\ then\ the\ derivative\ of\ f(x) \times g(x)\ is$

Select the correct option

- $x^2 + 12.$
- $12x^2 + 6x.$
- $6x^2 + 12x.$
- None of these.

Click to Save Answer &

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Question # 7 of 30 (Start time: 05:22:18 PM, 21 June 2021)

$What\ is\ the\ derivative\ of\ \cos ec(x^4)?$

Select the correct option

- $\cos ec(x^4)\cot(x^4)$
- $-\cos ec(x^4)\cot(x^4)$
- $None\ of\ these$
- $-4x^3\cos ec(x^4)\cot(x^4)$

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Quiz Start Time: 05:18 P

Question # 8 of 30 (Start time: 05:22:54 PM, 21 June 2021)

Which of the following rule will be used to find the derivative of $xcotx$?

Select the correct option

Reload

- sum
- product
- power
- difference

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Quiz Start Time

Question # 8 of 30 (Start time: 05:22:54 PM, 21 June 2021)

Which of the following rule will be used to find the derivative of $xcotx$?

Select the correct option

- sum
- product
- power
- difference

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Quiz Start Time:

Question # 9 of 30 (Start time: 05:24:02 PM, 21 June 2021)

$Let \;\; f(x) = x - 3 \;\; and \;\; g(x) = x^3 \;\; then \;\; (f-g)(x) = ?$

Select the correct option

○	$-x^3 - x - 3$
○	$x^3 + x - 3$
○	$-x^3 + x + 3$
●	$-x^3 + x - 3$

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Quiz Start Tim

Question # 10 of 30 (Start time: 05:24:32 PM, 21 June 2021)

$Derivative\ of\ \sin(u^5)\ is.....$

Select the correct option

●	$5u^4 \cos u^5$
○	$-5u^4 \cos u^5$
○	$5u^4 \sin u^5$
○	$-5u^4 \sin u^5$

Click to Save Answer &

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Question # 11 of 30 (Start time: 05:25:01 PM, 21 June 2021)

Which of the following describes the equation:
x = 1?

Select the correct option

- A point (1, 0)
- A line parallel to x-axis
- A line parallel to y-axis
- A point (0, 1)

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Quiz Start Tim

Question # 12 of 30 (Start time: 05:25:34 PM, 21 June 2021)

$If\ f(x) = x \sin x^2, then\ which\ of\ the\ following\ is\ true\ about\ it$

Select the correct option

- $Its\ derivative\ w.r.t\ x\ is\ 2x \cos x^2 + \sin x^2$
- $Its\ derivative\ w.r.t\ x\ is\ 2x^2 \cos x^2 + \sin x^2$
- $None\ of\ these$
- $Its\ derivative\ w.r.t\ x\ is\ x \cos x^2 + \sin x^2$

Click to Save Answer &

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Question # 13 of 30 (Start time: 05:26:01 PM, 21 June 2021)

The derivative of composition of functions can be found by using _____.

Select the correct option

- Power rule
- Generlized derivative formula
- None of these
- Chain rule

Click to

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Question # 14 of 30 (Start time: 05:26:32 PM, 21 June 2021)

For the curve f(t) = -23t+25, the slope of the tangent line at point 't = -1' is _____.

Select the correct option

- 23
- -25
- 25
- -23 (selected)

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Question # 15 of 30 (Start time: 05:26:58 PM, 21 June 2021)

The derivative of x^3+x^2 is _____.

Select the correct option

- 3x-2
- 3x^2+2x
- 3x+2
- 3x^2-2x

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Quiz Start Time:

Question # 16 of 30 (Start time: 05:27:28 PM, 21 June 2021)

$If\ f(x) = \cos x^5, then$ which of the following is true about it

Select the correct option

R

- None of these.
- *Its derivative w. r. t x is* $-5x^4 \sin x^5$
- *Its derivative w. r. t x is* $5x^4 \sin x^5$
- *Its derivative w. r. t x is* $-x^5 \sin x^4$

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Question # 17 of 30 (Start time: 05:27:54 PM, 21 June 2021)

The interval (-3, 2) denote the set of all _____between -3 and 2.

Select the correct option

- Integers
- Natural numbers
- Rational numbers
- Real numbers

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Question # 18 of 30 (Start time: 05:28:21 PM, 21 June 2021)

A function f(x) = x / |x| is

Select the correct option

- not continuous at x = 1
- continuous at x = 1

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Qu

Question # 19 of 30 (Start time: 05:28:50 PM, 21 June 2021)

Derivative of a graph of straight line bisecting the first and third quadrants is------

Select the correct option

- Infinity
- +1
- -1
- Zero

Click to Save A

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Question # 20 of 30 (Start time: 05:29:23 PM, 21 June 2021)

Which of the following is solution of an inequality:
$|x + 3| > 2$

Select the correct option

- x < -5 or x > -1
- x > -5 or x < -1
- x < -5 or x < -1
- x > -5 or x > -1

Click to S

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Question # 21 of 30 (Start time: 05:29:48 PM, 21 June 2021)

Which of the following is not an example of set?

Select the correct option

- Collection of prime numbers which are even.
- Collection of some stars in the sky.
- Collection of whole numbers between 20 and 25.
- Collection of colors in rainbow.

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Question # 22 of 30 (Start time: 05:30:16 PM, 21 June 2021)

If a function f has a relative extrema at a point c, then c is a critical point for f.

Select the correct option

- (●) True
- ( ) False

Cl

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Question # 23 of 30 (Start time: 05:30:45 PM, 21 June 2021)

The interval $[a, b)$ is defined by

Select the correct option

- $\{x : a \leqslant x < b\}$
- $\{x : a < x < b\}$
- $\{x : a \leqslant x \leqslant b\}$
- $\{x : a < x \leqslant b\}$

Click to S

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Question # 24 of 30 (Start time: 05:31:09 PM, 21 June 2021)

|-7| + |- 4| = _____

Select the correct option

○	-11
○	Does not exist
●	11
○	±11

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Question # 25 of 30 (Start time: 05:31:49 PM, 21 June 2021)

Which of the following is midpoint of the line segment joining points (1, 3) and (1, 5)?

Select the correct option

- (0, 2)
- (1, 4)
- (2, 8)
- (0, -1)

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Question # 26 of 30 (Start time: 05:32:24 PM, 21 June 2021)

The derivative of a variable with respect to itself is equal to .....

Select the correct option

- 2
- 8
- 0
- 1

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Question # 27 of 30 (Start time: 05:32:56 PM, 21 June 2021)

$If\ g(X) = 3x^2\ then\ g'(2)\ will\ be\ .....\ ?$

Select the correct option

- 18
- -12
- 15
- 12

Click to Save

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Question # 28 of 30 (Start time: 05:33:26 PM, 21 June 2021)

Let a function f is twice differentiable at a stationary point x_0, then f has relative maximum at x_0 if

Select the correct option

- (●) $f''(x_0) < 0$
- ( ) $f''(x_0) = 0$
- ( ) $f''(x_0) > 0$
- ( ) $f''(x_0) \geq 0$

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Question # 29 of 30 (Start time: 05:33:50 PM, 21 June 2021)

$The\ derivative\ of\ (x^2) + 2sinx\ w.r.t\ x\ is$

Select the correct option

- -2x-2cosx
- 2x-2cosx
- -2x+2sinx
- 2x+2cosx

Click

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Question # 30 of 30 (Start time: 05:34:24 PM, 21 June 2021)

The derivative of (1/sinx) is .....

Select the correct option

- [x] -Cosecx.Cot(x)
- [ ] Cosecx.Cot(x)
- [ ] Sec(x)
- [ ] Cosec(x)

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Question # 8 of 30 (Start time: 05:55:47 PM, 21 June 2021)

What is the derivative of $\sin(x^{\pi})$?

Select the correct option

○	$-\pi(x)^{\pi-1}\cos(x^{\pi})$
●	$\pi(x)^{\pi-1}\cos(x^{\pi})$
○	$\cos(x^{\pi})$
○	None of these

Click to

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Quiz Start Time: 01:21 P

Question # 9 of 30 (Start time: 05:56:16 PM, 21 June 2021)

$Which\ of\ the\ following\ is\ VERTEX\ of\ parabola\ represented\ by\ the\ equation: y = x^2 + 2$

Select the correct option

Reload M

- (1, 3)
- (2, 6)
- (0, 2)
- None of these

Click to Save Answer & Move to N

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Question # 10 of 30 (Start time: 05:56:52 PM, 21 June 2021)

d(5x+4)/dx=.....

Select the correct option

○	2
○	9
○	4
●	5

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Question # 11 of 30 (Start time: 05:57:19 PM, 21 June 2021)

$2x + 3y = 6$ is .....

Select the correct option

- A function
- Not a function

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Question # 12 of 30 (Start time: 05:57:43 PM, 21 June 2021)

The implicit differentiation of $y^2 = x$, gives $\frac{dy}{dx} =$ _____.

Select the correct option

- $\frac{1}{2y}$
- 1
- $2y\frac{dy}{dx}$
- $\frac{1}{2x}$

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Question # 13 of 30 (Start time: 05:58:06 PM, 21 June 2021)

If xy=3, then dy/dx=-3/x

Select the correct option

- (●) False
- ( ) True

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Question # 14 of 30 (Start time: 05:58:40 PM, 21 June 2021)

Which of the following is distance between the points (2, 1) and (1, 1)?

Select the correct option

- [x] 1
- [ ] Square root of (13)
- [ ] (1, 0)
- [ ] -1

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Quiz Start Time: 0

Question # 15 of 30 (Start time: 05:59:10 PM, 21 June 2021)

Derivative of $\sin(5x+3)$ *is* .....

Select the correct option

Rel

- $5cos(5x+3)$
- $-5\sin(5x+3)$
- $3\cos(5x+3)$
- $3\cos 5x$

Click to Save Answer & Move

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Question # 16 of 30 (Start time: 05:59:48 PM, 21 June 2021)

Which of the following describes the equation:
x = 1?

Select the correct option

- A point (1, 0)
- A line parallel to y-axis
- A point (0, 1)
- A line parallel to x-axis

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Question # 17 of 30 (Start time: 06:00:39 PM, 21 June 2021)

Which of the following is distance between the points (1, 2) and (1, 3)?

Select the correct option

- [x] 1
- [ ] (0, -1)
- [ ] (2, 5)
- [ ] 0

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Quiz Star

Question # 18 of 30 (Start time: 06:01:01 PM, 21 June 2021)

$If\ f(x) = x^5\ and\ g(x) = x^2,\ then\ the\ derivative\ of\ \frac{f(x)}{g(x)}\ is$

Select the correct option

- 3x
- $3x^2$
- None of these
- $\frac{3}{x^2}$

Click to Save Answe

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Question # 19 of 30 (Start time: 06:01:28 PM, 21 June 2021)

The derivative of sec(3x) is

Select the correct option

- 3Sec3x.tanx
- Sec3x.tan3x
- 3.Sec3x.tan3x
- 3Secx.tanx

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Question # 20 of 30 (Start time: 06:01:57 PM, 21 June 2021)

If for any positive number e(epsilon) we can find d (delta) such that| (3x-5) - 1| < e ,if x satisfies
0< |x-2| < d
Then f(x) =.....

Select the correct option

- None of these
- 3x-5
- x-2
- 3x-5- 1

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Question # 21 of 30 (Start time: 06:02:29 PM, 21 June 2021)

Which of the following rule will be used to find the derivative of $xcotx$?

Select the correct option

- difference
- sum
- product
- power

Click to Save

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Question # 22 of 30 (Start time: 06:02:59 PM, 21 June 2021)

$$\lim\left[\frac{f(x)}{g(x)}\right] = \frac{\lim f(x)}{\lim g(x)}; \quad \lim g(x) \text{ is}$$

Select the correct option

- ○ Equal to zero
- ● Not equal to zero
- ○ One
- ○ Undefined

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Question # 23 of 30 (Start time: 06:03:21 PM, 21 June 2021)

Velocity is the rate of change of position w.r.t .....

Select the correct option

- (●) time
- ( ) force
- ( ) acceleration
- ( ) displacement

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Question # 24 of 30 (Start time: 06:03:48 PM, 21 June 2021)

$If\ \mathrm{f(x)} = \mathrm{cosx}^5, then$ which of the following is true about it

Select the correct option

- $Its\ derivative\ w.r.t\ x\ is\ -5x^4 \sin x^5$
- $Its\ derivative\ w.r.t\ x\ is\ 5x^4 \sin x^5$
- None of these.
- $Its\ derivative\ w.r.t\ x\ is\ -x^5 \sin x^4$

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Question # 25 of 30 (Start time: 06:04:11 PM, 21 June 2021)

If $x - y$ is positive, then

Select the correct option

- $y > x$
- $x = y$
- $x > y$
- $x < y$

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Question # 26 of 30 (Start time: 06:04:39 PM, 21 June 2021)

The function f is said to be _____ on (a, b), if $f''(x) > 0$ on that interval.

Select the correct option

- Concave down
- Constant
- Concave up
- None of these

Click to Sav

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Question # 27 of 30 (Start time: 06:05:08 PM, 21 June 2021)

A function .....be differentiable at the point of discontinuity.

Select the correct option

- cannot
- can

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Question # 28 of 30 (Start time: 06:05:31 PM, 21 June 2021)

At a corner a tangent line does not exist because the slopes of the secant lines do not have a (two sided) limit.

Select the correct option

- (●) True
- ( ) False

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Question # 29 of 30 (Start time: 06:05:54 PM, 21 June 2021)

Which of the following point satisfies the equation: y = -3x + 2

Select the correct option

- (-1, 4)
- (-3, 2)
- (1, 4)
- (2, -4)

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Question # 30 of 30 (Start time: 06:06:25 PM, 21 June 2021)

The inequality L-e < f(x) < L+ e can be written as .....

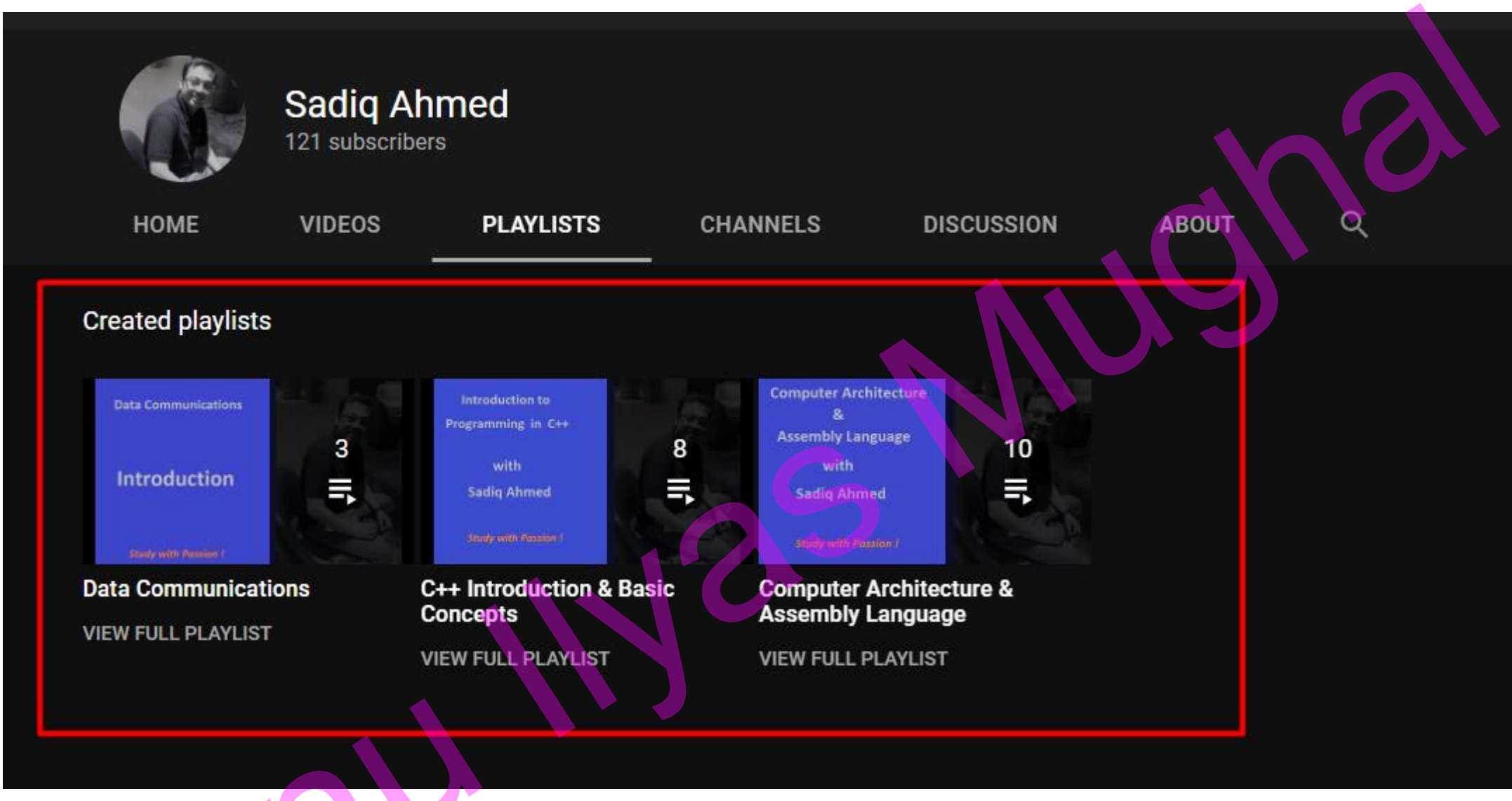
Select the correct option

- | f(x)- e | < L
- | f(x)- L | > e
- |f(x)- e | > L
- | f(x)- L | < e

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Best short lectures by Tutor Sadiq Ahmed

https://www.youtube.com/channel/UCjboEX65C_yN7as5di1OuIQ/playlists

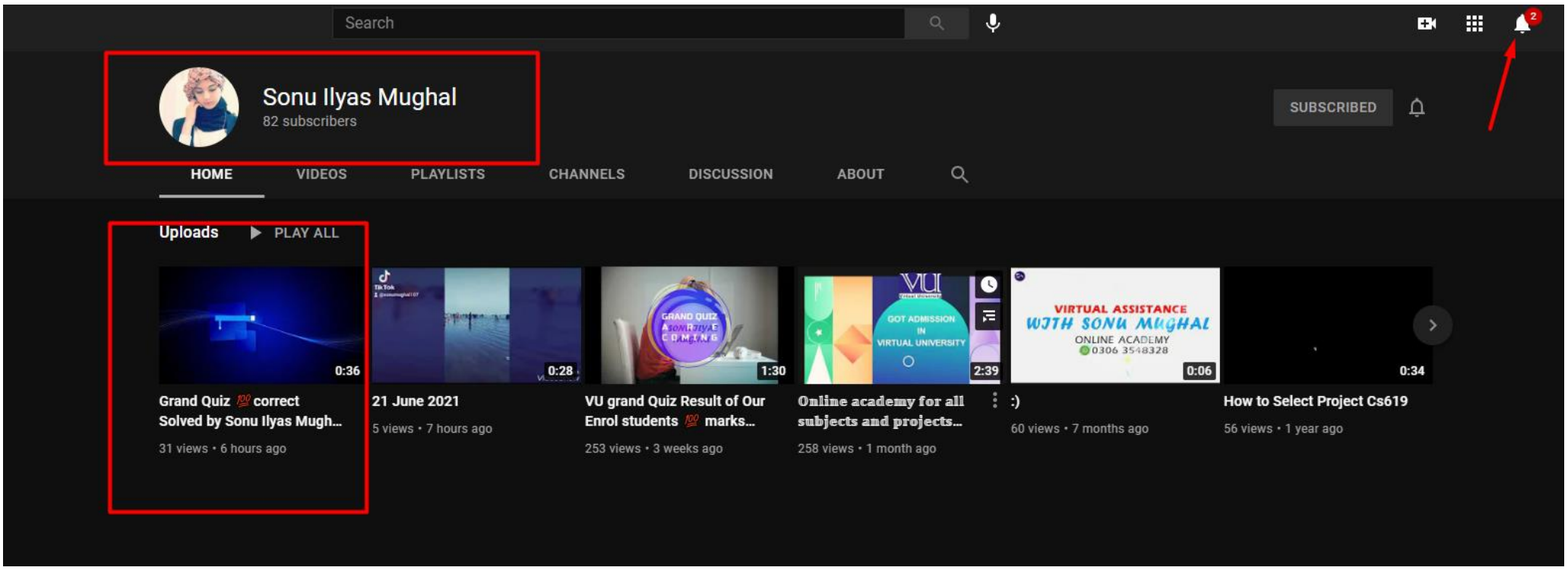


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3) Recorded Videos
4) Topic wise Notes
5) Short QA Solved
6) MCQs Solved
7) Solved Past and Current paper
8) Solved Graded Activities
(Assignment Quiz GDB)

Activity
Detailed
Information

Project Enrolment Complete training with
Live classes on Zoom
1) SRS (Software Requirements Specification)
2) DD (Design Document)
3) Test phase
a) Test phase viva (Live Classes for viva preparation)
4) Final Deliverable
a) Final coding
b) Final presentation
c) Final report
d) Pre final viva (Coding Viva)
e) Final viva (Overall Presentation)

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